

Curriculum Vitae

Charikleia Karageorgiou

Postdoctoral Researcher, Department of Biological Sciences, University at Buffalo

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 Google Scholar |  ORCID |  Bluesky |  Website

EDUCATION

- Sept 2017 – Sept 2022 PhD *cum laude*, Evolutionary Genetics, Universitat Autònoma de Barcelona, Spain.
- Sept 2016 – July 2017 MSc Bioinformatics, Universitat Autònoma de Barcelona, Spain.
- Sept 2014 – July 2015 MSc Advanced Genetics, Universitat Autònoma de Barcelona, Spain.
- Sept 2011 – June 2014 BSc (Hons) Biomedicine, University of East Anglia, United Kingdom.

ACADEMIC POSITIONS

- Jan 2023 – Present Postdoctoral Researcher, Dept. Biological Sciences, University at Buffalo, SUNY, NY, USA. Mentor: Dr. Omer Gokcumen
- Dec 2021 – July 2022 Research Fellow, Institute of Computer Science, Foundation for Research and Technology Hellas - Greece. Supervisor: Dr. Pavlos Pavlidis
- July 2015 – Dec 2016 Research Assistant, Grup de Genòmica, Bioinformàtica i Biologia Evolutiva, Universitat Autònoma de Barcelona, Spain.

VISITING POSITIONS & INTERNSHIPS

- Oct 2025 – Nov 2025 Visiting Researcher, Center for Interdisciplinary Theoretical and Mathematical Sciences, RIKEN, Japan. Mentor: Dr. Leo Speidel
- Aug 2021 – Feb 2022 Visiting PhD Researcher, Institute of Molecular Biology and Biotechnology, Foundation for Research and Technology Hellas, Greece. Funded by Marie Skłodowska-Curie Actions, Research and Innovation Staff Exchange (RISE). Supervisor: Dr. John Vontas
- Feb 2021 – June 2021 PhD Internship, University of Zürich, Switzerland. Supervisor: Dr. Martin Kapun

HONORS & AWARDS

- 2025 DeLill Nasser Award for Professional Development in Genetics, Genetics Society of America. (\$1,000)
- 2022 *Cum Laude*, PhD Defense, Universitat Autònoma de Barcelona, Spain.
- 2020 Swiss-European Mobility Programme (SEMP) Grant (440 CHF/month, declined due to COVID-19 pandemic)
- 2017 – 2022 PIF PhD fellowship from the Universitat Autònoma de Barcelona, Spain. (~75,000 €)

PUBLICATIONS & PREPRINTS

* denotes co-first authorship, # denotes corresponding author

12. **Karageorgiou, C.**[#], Leffler, E.M.[#], Dennis, M.Y.[#] and Gokcumen, O.[#], 2026. Structural variation in context: mechanisms, functions and selection regimes across the tree of life. *Heredity*. [doi: <https://doi.org/10.1038/s41437-026-00832-x>]
11. Aguado, W.D., **Karageorgiou, C.**, Kassem, S., Hernandez Rojas, R., Veilleux, C.C., Melin A.D., Vogel, E.R., Ruhl, S., Gokcumen O.[#] and Pajic P.[#], 2026. Multiscale structural evolution shapes primate salivary defenses. *Under review (MBE)*.
10. Acharjee, M., Humphrey, K., Lactaoen, V., **Karageorgiou, C.**, Liu, B., Gokcumen, O. and Rusche, L.[#]. 2026. The deacetylase Sir2 is the primary sensor driving transcriptional changes in response to low NAD⁺ in the yeast *Kluyveromyces lactis*. *Genetics*, iyag025. [doi: <https://doi.org/10.1093/genetics/iyag025>]
9. **Karageorgiou, C.**, Pajic, P., Ruhl S. and Gokcumen, O.[#], 2026. Convergent evolution through independent rearrangements in the primate amylase locus. *In revision*. [doi: <https://doi.org/10.1101/2025.08.14.670395>]
8. Scheer, K.*[#], Landau, L.J.B.*[#], Jorgensen, K., **Karageorgiou, C.**, Siao, L., Alkan, C., Morales-Rivera, A.M., Osbourne, C., Garcia, O., Pearson, L., Kiyamu, M., Rivera-Chira, M., Leon-Velarde, F., Lee, F., Brutsaert, T., Bigham, A.W.[#] and Gokcumen, O.[#], 2025. Rapid Adaptive Increase of Amylase Gene Copy Number in Indigenous Andeans. *In Press (Nature Communications)*. [doi: <https://doi.org/10.1101/2025.03.25.644684>]
7. Running, L.*[#], Cristobal, J.R.*[#], **Karageorgiou, C.**, Camdzic, M., Aguilar, J.M.N., Gokcumen, O., Aga, D.S.[#] and Atilla-Gokcumen, G.E.[#], 2024. Investigating the Mechanism of Neurotoxic Effects of PFAS in Differentiated Neuronal Cells through Transcriptomics and Lipidomics Analysis. *ACS Chemical Neuroscience*, 15(24), pp.4568-4579. [doi: <https://doi.org/10.1021/acscchemneuro.4c00652>]
6. Yilmaz, F.*[#], **Karageorgiou, C.*[#]**, Kim, K.*[#], Pajic, P., Scheer, K., Human Genome Structural Variation Consortium, Beck, C.R., Torregrossa, A.M., Lee, C.[#] and Gokcumen, O.[#], 2024. Reconstruction of the human amylase locus reveals ancient duplications seeding modern-day variation. *Science*, 386(6724), p.eadn0609. [doi: <https://doi.org/10.1126/science.adn0609>]
5. Papadadonakis, S., Kioukis, A., **Karageorgiou, C.**, and Pavlidis, P.[#] 2024. Evolution of gene regulatory networks by means of selection and random genetic drift. *PeerJ*, 12, e17918. [doi: <https://doi.org/10.7717/peerj.17918>]
4. **Karageorgiou, C.**, Gokcumen, O. and Dennis, M.[#], 2024. Deciphering the Role of Structural Variation in Human Evolution: A Functional Perspective. *Current Opinion in Genetics & Development*, 88, p.102240. [doi: <https://doi.org/10.1016/j.gde.2024.102240>]
3. Labbé, F., et al., 2023. Genomic analysis of two phlebotomine sand fly vectors of *Leishmania* from the New and Old World. *PLoS Neglected Tropical Diseases*, 17(4), p.e0010862. [doi: <https://doi.org/10.1371/journal.pntd.0010862>]
2. **Karageorgiou, C.**[#], Tarrío, R.[#] and Rodríguez-Trelles, F.[#], 2020. The Cyclically Seasonal *Drosophila subobscura* Inversion O₇ Originated From Fragile Genomic Sites and Relocated Immunity and Metabolic Genes. *Frontiers in genetics*, 11, p.1190. [doi: <https://doi.org/10.3389/fgene.2020.565836>]
1. **Karageorgiou, C.**[#], Gámez-Visairas, V., Tarrío, R.[#] and Rodríguez-Trelles, F.[#], 2019. Long-read based assembly and synteny analysis of a reference *Drosophila subobscura* genome reveals signatures of

structural evolution driven by inversions recombination-suppression effects. *BMC Genomics*, 20(1), p.1-21. [doi: <https://doi.org/10.1186/s12864-019-5590-8>]

TEACHING & MENTORSHIP

Mentoring:

▫ Carsyn Bonesteel, University at Buffalo, 2024 – Present. BSc Student. Mentoring her independent research project on the evolutionary and functional impact of the hominin-specific *ACOT1* gene duplication on metabolic health.

Awards under supervision:

- ▶ Best Idea Award, Department of Biological Sciences, December 2024
- ▶ Gerace Dean's Innovation Fund Award, May 2025
- ▶ SUNY Summer Undergraduate Research Program (SURP), paid research internship, Summer 2025
- ▶ McCroskey Fellowship, October 2025
- ▶ UB nominee for the Goldwater Scholarship, December 2025
- ▶ GSA Undergraduate Travel Award, January 2026
- ▶ Summer Research Experience for Undergraduates (REU), Johns Hopkins University, Summer 2026

▫ Rebekah Sheih, University at Buffalo, Summer 2024. BSc REU Student. Supervised her REU project on characterizing the evolutionary history of the *ACOT1* locus.

Currently: PhD Student at UConn Health.

▫ Eliseo Pampín Bello, Universitat Autònoma de Barcelona, 2017 – 2018. Mentored MSc student and supervised MSc Thesis “Genomic localization and evolutionary history of two aminopeptidase genes long-known to exhibit systematic patterns of spatiotemporal variation in natural populations of *Drosophila*”.

Teaching:

BIO550LEC – Human Evolutionary Genomics, Graduate Course, Department of Biological Sciences, University at Buffalo, Fall 2025. (1 Lecture)

101963 – Genetics, BSc Genetics, Faculty of Biosciences, Universitat Autònoma de Barcelona, 2017 – 2021.

Delivered lectures and served as a teaching assistant for courses covering *Drosophila morphology*, *Genetic linkage & mapping*, *Recombination*, and *Population Genetics*.

APPOINTMENTS & EDITORIAL ROLES

Editor: Associate Editor for *Heredity* (2025 – Present).

Guest-Editor: *Heredity*, Special Issue: *Functional and Adaptive Effects of Genomic Structural Variation*, (with Omer Gokcumen, Megan Dennis, and Ellen Leffler). 2024 – 2025.
<https://www.nature.com/collections/deejeiccee>

INVITED SEMINARS

Mar 2026 Why complexity persists: Evolutionary dynamics of the amylase locus in humans and non-human primates, *Invited Seminar*, Malaspinas Lab, University of Lausanne, Switzerland.

Mar 2026 Origins of recurrent structural variation: lessons from the primate amylase locus, *Invited Seminar*, Flatt Lab, University of Fribourg, Switzerland.

Dec 2025 Origins of recurrent structural variation: lessons from the primate amylase locus, *Invited Seminar*, Department of Evolutionary Anthropology, University of Vienna, Austria.

- Oct 2025 Why complexity persists: Evolutionary dynamics of the amylase locus in primates, **Invited Seminar**, Biology Seminar, iTHEMS, RIKEN, Japan.
- Mar 2025 Reconstruction of the human amylase locus reveals ancient duplications seeding modern-day variation, **Invited Seminar**, Brock University, St Catharines, Canada.
- Feb 2024 Ancient *AMY1* gene duplications primed the amylase locus for adaptive evolution upon the onset of agriculture, **Invited Seminar**, Biological Sciences Seminar Series, University at Buffalo, NY, USA.
- Mar 2022 Comparative population genomics of natural populations of *Drosophila subobscura*: Understanding the establishment and maintenance of chromosomal inversion polymorphisms. **Invited Seminar**, The Eric Lai Lab, Sloan Kettering Institute, NY, USA.
- Feb 2022 Comparative population genomics of natural populations of *Drosophila subobscura*: Understanding the establishment and maintenance of chromosomal inversion polymorphisms. **Invited Seminar**, Gokcumen Lab, University at Buffalo, NY, USA.
- Sept 2021 Population genomics of inversion polymorphisms in *Drosophila subobscura*: Mechanisms of establishment and maintenance, **Invited Seminar**, Pavlidis Lab, FORTH, Crete, Greece.

CONFERENCE CONTRIBUTIONS

- 2026 Integrating long-read assemblies and ancestry graphs to map Neanderthal-introgressed structural variation in humans, **Poster**, PEQG 2026, Pacific Grove, CA, USA.
- 2025 Convergent evolution through recurrent rearrangements and regulatory rewiring in the primate amylase locus, **Talk**, ESEB 2025, Barcelona, Spain.
- 2025 Reconstruction of the human amylase locus reveals ancient duplications seeding modern-day variation, **Poster**, ESEB 2025, Barcelona, Spain.
- 2025 Why this locus? Recurrent duplications and extensive structural polymorphism at the amylase locus in humans and non-human primates, **Talk**, Ecological and Evolutionary Genomics, Gordon Research Conference, Renaissance Tuscany Il Ciocco, Italy.
- 2025 Convergent evolution through recurrent rearrangements in the primate amylase locus, **Talk**, Evolution 2025, Athens, GA, USA.
- 2025 Reconstruction of the human amylase locus reveals ancient duplications seeding modern-day variation, **Poster**, Evolution 2025, Athens, GA, USA.
- 2025 Reconstruction of the human amylase locus reveals ancient duplications seeding modern-day variation, **Talk**, Virtual Evolution 2025, Athens, GA, USA.
- 2025 Convergent evolution through recurrent rearrangements in the primate amylase locus, **Talk**, Virtual Evolution 2025, Athens, GA, USA.
- 2024 Reconstruction of the human amylase locus reveals ancient duplications seeding modern-day variation, **Poster**, Annual Postdoctoral Research Symposium, University at Buffalo, NY, USA.
- 2024 Reconstruction of the human amylase locus reveals ancient duplications seeding modern-day variation, **Talk**, GLAM EvoGen 2024, Syracuse University, NY, USA.
- 2024 Paleolithic gene duplications primed adaptive evolution of human amylase locus upon agriculture, **Poster**, SBE 2024, Puerto Vallarta, Mexico.

- 2023 Modeling the evolutionary and functional impact of genomic structural variation in mammalian saliva. **Poster**, Ecological and Evolutionary Genomics, Gordon Research Conference, Bryant University, RI, USA.
- 2022 Disentangling the origin of the Cretan wild goat *Capra aegagrus cretica*. **Talk**, 15th International Congress on the Zoogeography and Ecology of Greece and Adjacent Regions, Mytilene, Greece.
- 2022 Disentangling the origin of the Cretan wild goat *Capra aegagrus cretica*. **Talk**, Third Joint NEOMATRIX Meeting, Virtual.
- 2021 The cyclically seasonal *Drosophila subobscura* inversion *O*₇ originated from fragile genomic sites and relocated immunity and metabolic genes. **Poster**, SMBE 2021. Virtual.
- 2021 The cyclically seasonal *Drosophila subobscura* inversion *O*₇ originated from fragile genomic sites and relocated immunity and metabolic genes. **Poster**, SMRT Leiden 2021 – Young Investigator Virtual Conference. Virtual.
- 2019 Long-read based assembly and synteny analysis of a reference *Drosophila subobscura* genome reveals signatures of structural evolution driven by inversions recombination-suppression effects. **Poster**, VII Jornada de Bioinformatica i Genomica, Barcelona, Spain.
- 2019 Long-read based assembly and synteny analysis of a reference *Drosophila subobscura* genome reveals signatures of structural evolution driven by inversions recombination-suppression effects. **Poster**, European Drosophila Research Conference (EDRC), Lausanne, Switzerland.
- 2019 Comparative population genomics of natural populations of *Drosophila subobscura*. **Talk**, VII Jornades de Biorecerca UABio. Universitat Autònoma de Barcelona. Barcelona, Spain.
- 2018 The genome of *Drosophila subobscura*. **Talk**, VI Jornades de Biorecerca UABio. Universitat Autònoma de Barcelona, Barcelona, Spain.

MEETINGS, ORGANIZATION & OUTREACH

- August 2025 Outreach Organizer – *Breaking Bread: Exploring the Evolution of Amylase in Humans*. Evolution, Diversity, Art and Community, Festa Major de Gràcia & ESEB 2025, Barcelona, Spain. Organized and led a bilingual (English/Spanish) public outreach activity on human evolution and the amylase locus, selected as part of ESEB’s outreach initiative. Co-organizer: Thomas Hitchcock.
- July 2025 Co-Chair, GRC Power Hour™ – Gordon Research Conference (GRC) on Evolutionary and Ecological Genomics: *Dynamics of Ecological and Evolutionary Change*, Lucca, Italy.
- July 2025 Chair for the Gordon Research Seminar (GRS) Ecological and Evolutionary Genomics: *Elucidating the Evolutionary Dynamics of Adaptation in Fluctuating Environments*, Lucca, Italy. Co-chair: Joaquin Nunez.
- May 2025 Moderator, Virtual Evolution 2025 Conference, Sessions: *Population Genetics Theory and Molecular Evolution*
- May 2025 Organization Committee for *Paving Pathways: A multi-panel symposium highlighting scientists in various post-graduate biomedical careers*, Buffalo, USA.
- 2025 – Present Organizer of “Talk Nerdy to Me”, a monthly event dedicated to public speaking and science communication, aimed at making research accessible to general audiences.

- July 2024 Organizer for Society for Molecular Biology and Evolution 2024 Symposium: *Deciphering the functional and adaptive effects of genomic structural variation*, Puerto Vallarta, Mexico. Co-organizers: Megan Dennis, Omer Gokcumen, and Ellen Leffler.
- 2024 – Present Organizer, Postdoctoral Meetups, University at Buffalo, USA – Initiated and coordinated regular gatherings to build a supportive community, foster open communication and encourage interdisciplinary exchange.
- 2022 – Present Organizer for [Genome Structure, Evolution and Anthropology Seminar Series](#), Buffalo Evolutionary and Anthropological Genomics Lab, University at Buffalo, USA.

SERVICE & MEMBERSHIPS

- Reviewer:* Cell, Cell Genomics, Communications Biology, Genetics, Genome Research, Molecular Biology and Evolution, NAR Genomics and Bioinformatics, Nature Communications, Science Advances, The American Journal of Human Genetics.
- Grant Reviewer:* The Leakey Foundation, NordForsk (Nordic Council of Ministers).
- Professional Training:* GSA Journals Peer Review Training Program, Winter 2025. eLife Early-Career Reviewers Pool, Spring 2025.
- Memberships:* Genetics Society of America (GSA), Society for Molecular Biology and Evolution (SMBE), Society for the Study of Evolution (SSE), European Society for Evolutionary Biology (ESEB), The Genetics Society (GS).
- Committees:* Selection Committee for the Distinguished Postdoc Mentor Award 2025-2026, University at Buffalo, USA.
- Departmental Service:* Poster Grader, Biological Sciences Research Symposium 2025, University at Buffalo, USA.
Poster Grader, Biological Sciences Research Symposium 2026, University at Buffalo, USA.

MEDIA HIGHLIGHTS & PRESS COVERAGE

- 2026 Invited expert commentary on “[Archaeogenetic insights into the demographic history of Late Neanderthals](#)” (PNAS), featured in [National Geographic](#).
- 2024 Yilmaz, Karageorgiou et al. Reconstruction of the human amylase locus reveals ancient duplications seeding modern-day variation. [Press release](#). Highlights: [NYT](#), [Discover](#), [CNN](#), [NBC](#). Highlighted in the [NIH Director's blog](#), as review highlight in [Cell Genomics](#), [PLOS SciComm](#) “Top Stories in Human Evolution of 2024”, Leakey Foundation Podcast “Top Human Origins Discoveries of 2024”.

PROGRAMMING LANGUAGES

Bash, Python, R, MySQL, HTML, Perl

REFERENCES

- Omer Gokcumen <omergokc@buffalo.edu> ; Professor, University at Buffalo ; Postdoc Mentor.
- Charles Lee <Charles.Lee@jax.org> ; The Robert Alvine Family Chair and Professor, The Jackson Laboratory for Genomic Medicine ; Collaborator and co-author.

- ▶ Megan Dennis <mydennis@ucdavis.edu> ; *Professor, Biochemistry & Molecular Medicine, M.I.N.D. Institute, Genome Center, University of California Davis* ; Collaborator and co-author.
- ▶ Stefan Ruhl <shruhl@buffalo.edu> ; *Professor and Chair of Oral Biology, University at Buffalo School of Dental Medicine* ; Collaborator and co-author.

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